



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Date 31.05.26



Outline

Executive
Summary

Introduction

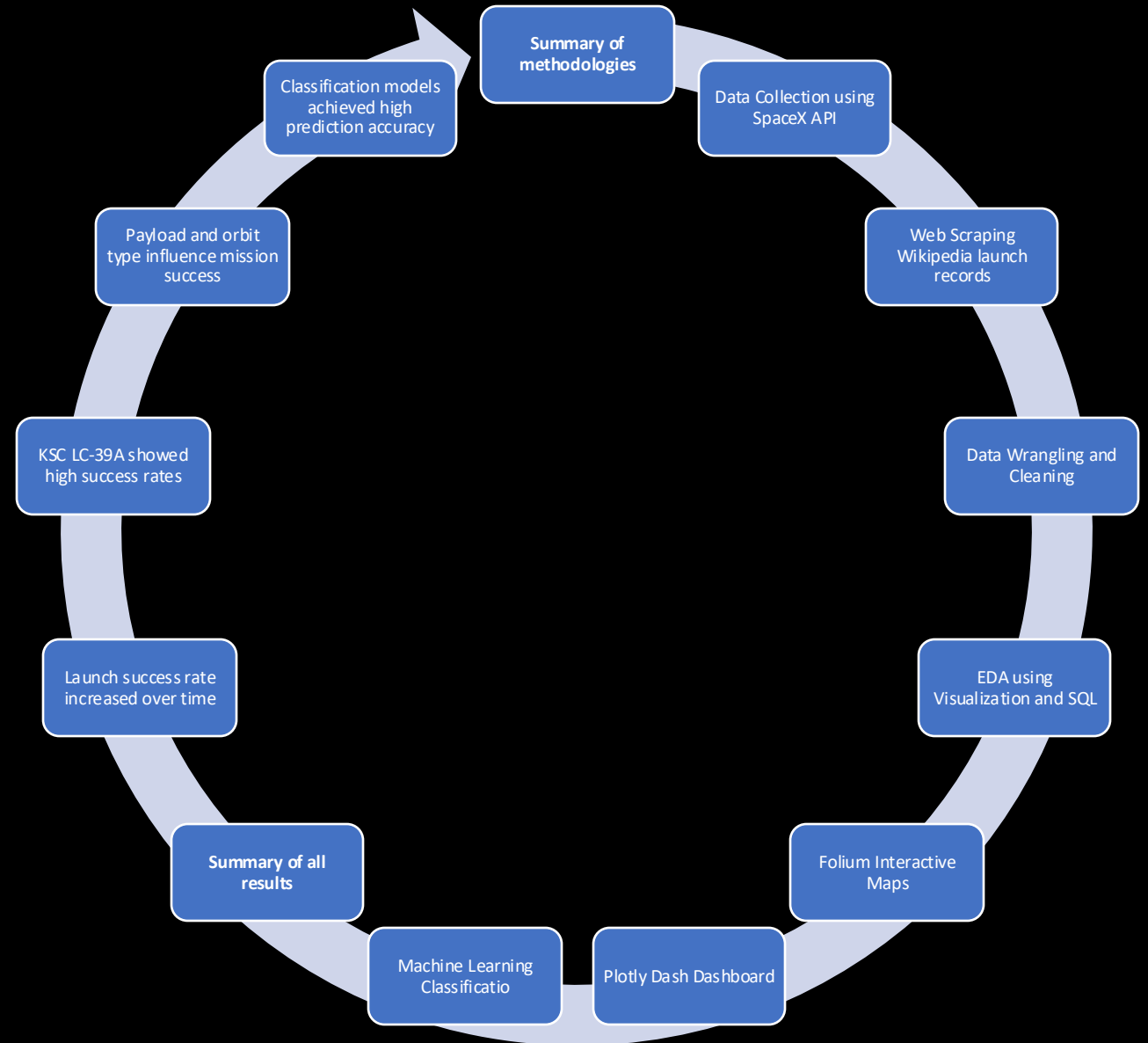
Methodology

Results

Conclusion

Appendix

Executive Summary



Introduction

Background

SpaceX significantly reduced launch costs through reusable rocket technology.

Problem Statement

Predict whether the Falcon 9 first stage will successfully land based on launch characteristics.

Business Value

Accurate predictions help estimate launch costs and mission success.

Section 1

Methodology

Methodology

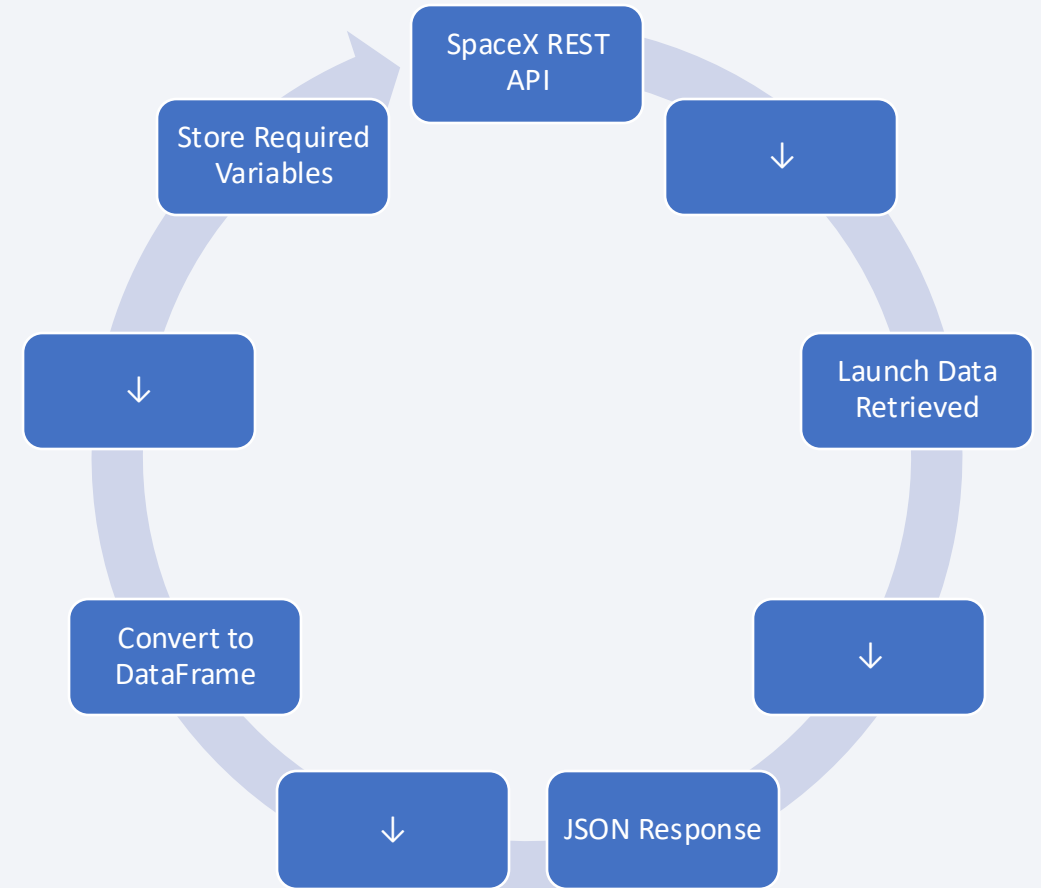
- Executive Summary
- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- Describe how data sets were collected.
- You need to present your data collection process use key phrases and flowcharts

Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose



Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts
- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose

Wikipedia Launch Records



BeautifulSoup



HTML Table Extraction



Convert to DataFrame



Store Launch Information

Data Wrangling

Activities

Removed missing values

Handled null landing outcomes

Created binary Class column

Standardized features

Generated final cleaned dataset

EDA with Data Visualization

Visualizations
Used

Scatter Plot

Bar Chart

Line Chart

Correlation
Analysis

Purpose

Understand
relationships
between:

Launch Site

Orbit

Payload

Flight Number

Mission
Success

EDA with SQL

SQL Queries
Performed

Unique launch
sites

Launch site
names starting
with CCA

Total payload
mass

Average
payload mass

Successful
ground
landings

Drone ship
landings

Mission
outcome
counts

Landing
outcome
rankings

Build an Interactive Map with Folium

- Objects Added
- Launch Site Markers
- Marker Clusters
- Success/Failure Colors
- Distance Lines
- Coastline Markers
- Railway Markers
- Highway Markers

Build a Dashboard with Plotly Dash

Dashboard Components



```
graph TD; A[Dashboard Components] --> B[Launch Site Dropdown]; B --> C[Success Pie Chart]; C --> D[Payload Range Slider]; D --> E[Payload vs Success Scatter Plot];
```

Launch Site Dropdown

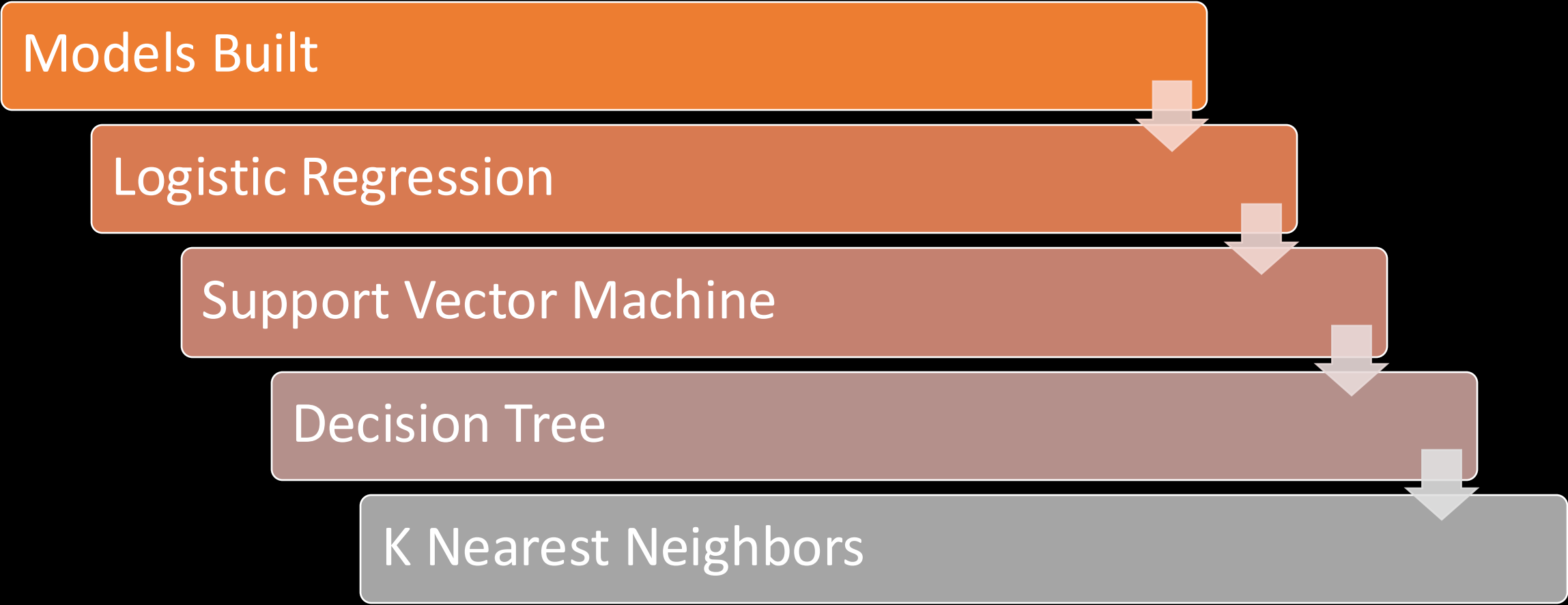
Success Pie Chart

Payload Range Slider

Payload vs Success Scatter Plot

Predictive Analysis (Classification)

Models Built



```
graph TD; A[Models Built] --> B[Logistic Regression]; B --> C[Support Vector Machine]; C --> D[Decision Tree]; D --> E[K Nearest Neighbors];
```

Logistic Regression

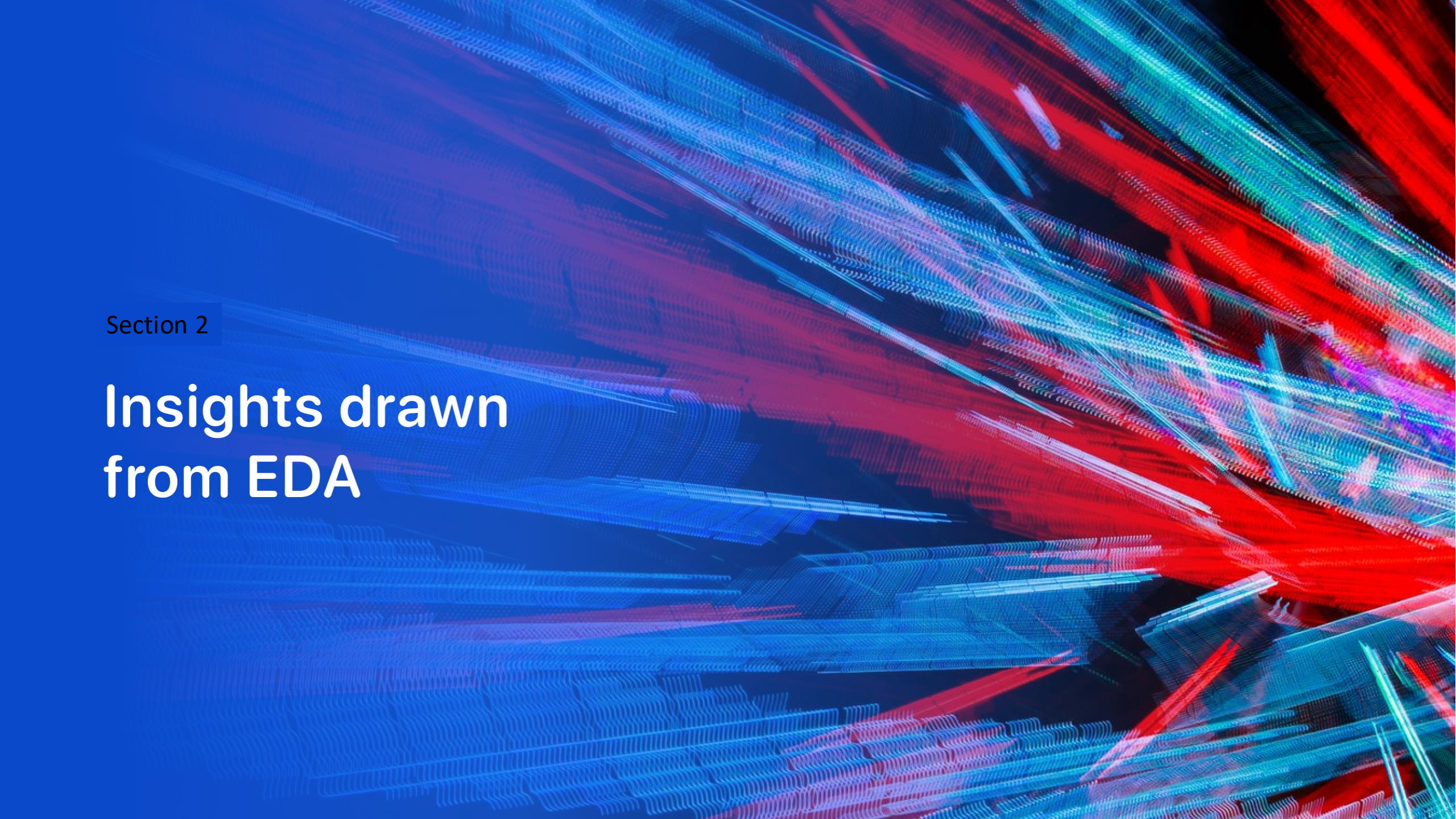
Support Vector Machine

Decision Tree

K Nearest Neighbors

Results

- Process
 - Data Preparation
 - ↓
 - Feature Scaling
 - ↓
 - Train-Test Split
 - ↓
 - Hyperparameter Tuning
 - ↓
 - Model Evaluation
 - ↓
 - Best Model Selecti

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

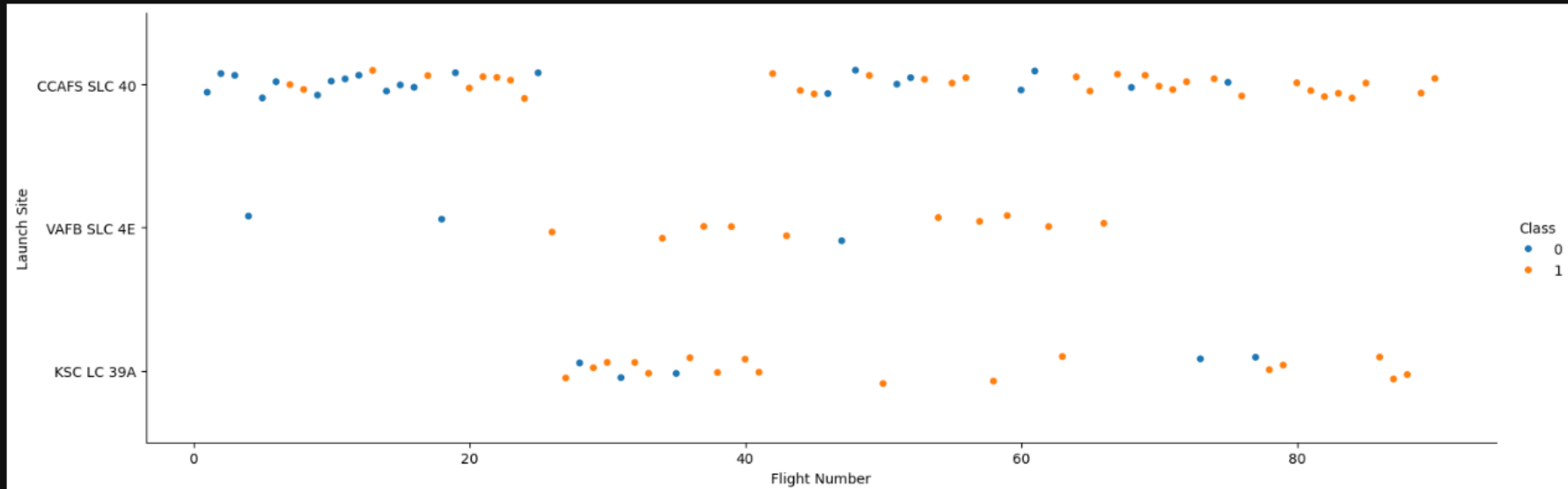
Insights drawn from EDA

Flight Number vs. Launch Site

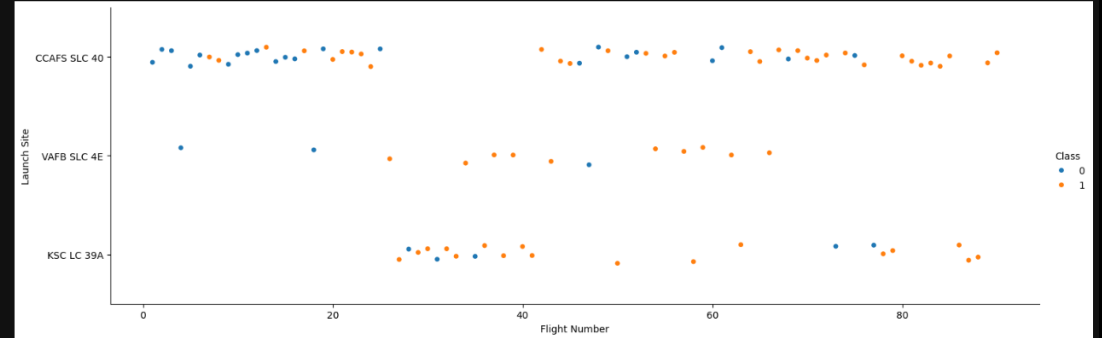
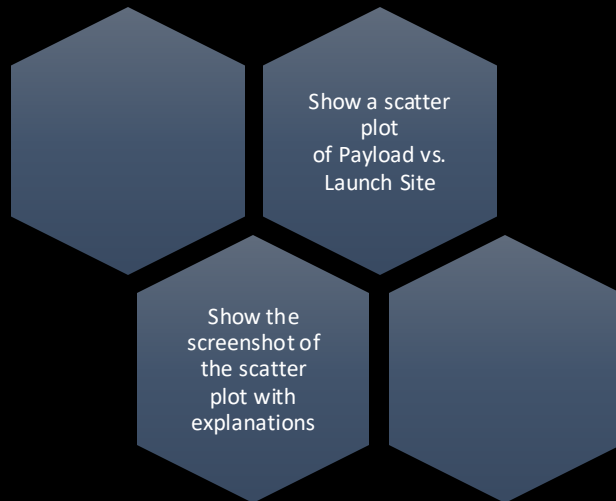
- Show a scatter plot of Flight Number vs. Launch Site
- Launch success increases as flight numbers increase, indicating improved reliability over time.

- Show the screenshot of the scatter plot with explanations

```
[7]: # Plot a scatter point chart with x axis to be Flight Number and y axis to be the launch site, and hue to be the class value
sns.catplot(y="LaunchSite", x="FlightNumber", hue="Class", data=df, aspect=3)
plt.xlabel("Flight Number")
plt.ylabel("Launch Site")
plt.show()
```



Payload vs. Launch Site



```
# HINT use groupby method on Orbit column and get the mean of Class column
```

```
success_rate = df.groupby('Orbit')['Class'].mean().reset_index()
```

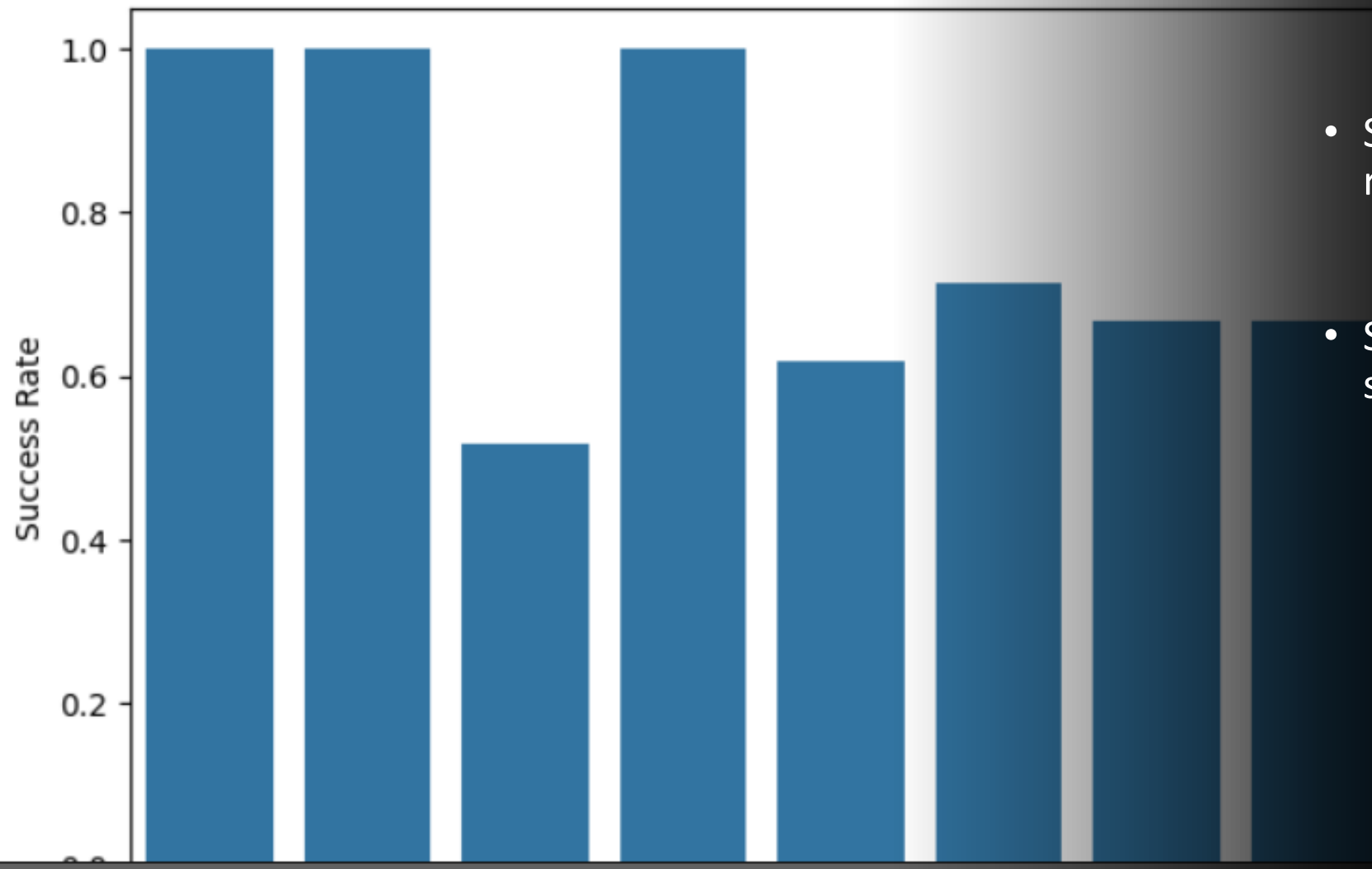
```
plt.figure(figsize=(10,5))
```

```
sns.barplot(x='Orbit', y='Class', data=success_rate)
```

```
plt.ylabel("Success Rate")
```

```
plt.show()
```

Success Rate vs. Orbit Type

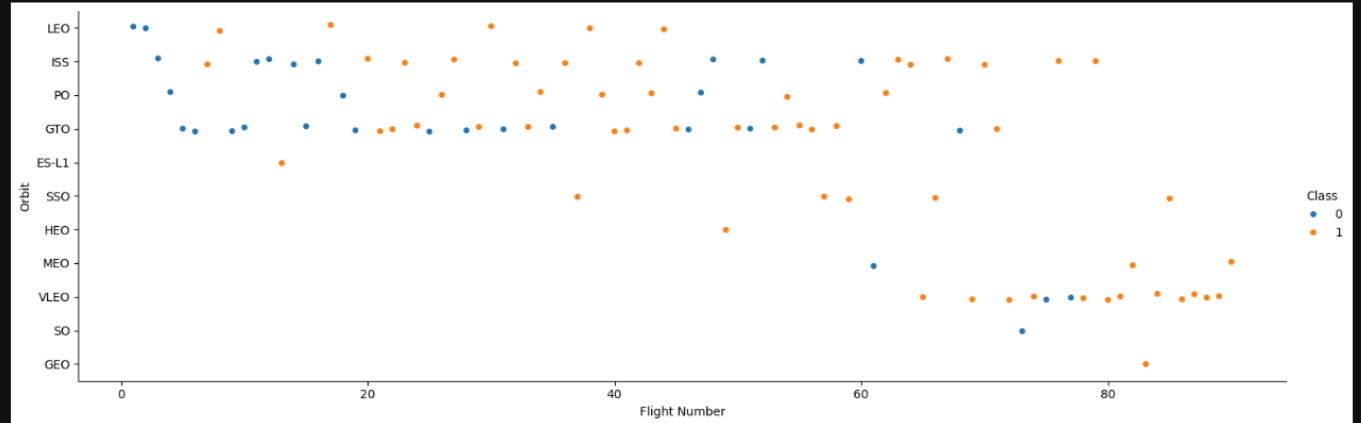


- Show a bar chart for the success rate of each orbit type
- Show the screenshot of the scatter plot with explanations

Flight Number vs. Orbit Type

- Show a scatter point of Flight number vs. Orbit type
- Show the screenshot of the scatter plot with explanations

```
[10]: # Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value
sns.catplot(y="Orbit", x="FlightNumber", hue="Class", data=df, aspect=3)
plt.xlabel("Flight Number")
plt.ylabel("Orbit")
plt.show()
```

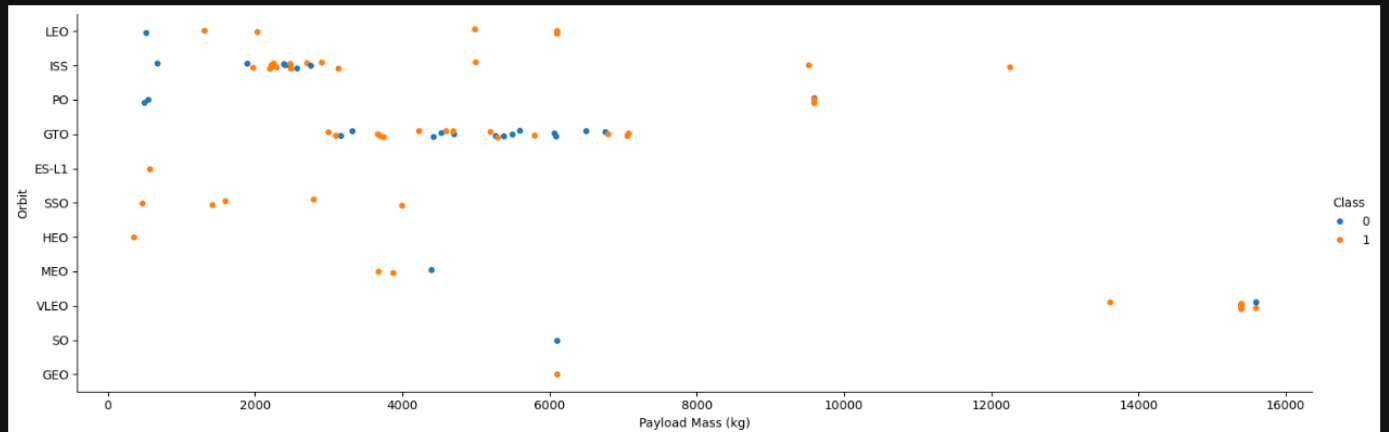


Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type
- Show the screenshot of the scatter plot with explanations

Similarly, we can plot the Payload Mass vs. Orbit scatter point charts to reveal the relationship between Payload Mass and Orbit type

```
[11]: # Plot a scatter point chart with x axis to be Payload Mass and y axis to be the Orbit, and hue to be the class value
sns.catplot(y="Orbit", x="PayloadMass", hue="Class", data=df, aspect=3)
plt.xlabel("Payload Mass (kg)")
plt.ylabel("Orbit")
plt.show()
```



Launch Success Yearly Trend

- Show a line chart of yearly average success rate
- Show the screenshot of the scatter plot with explanations

```
# A function to Extract years from the date
year=[]
def Extract_year():
    for i in df["Date"]:
        year.append(i.split("-")[0])
    return year
Extract_year()
df['Date'] = year
df.head()
```

er	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	La
1	2010	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.5
2	2012	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.5
3	2013	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.5
4	2013	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.6
5	2013	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.5

[5]:	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	BL
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	

All Launch Site Names

- Find the names of the unique launch sites
- Present your query result with a short explanation here

```
SELECT *  
FROM SPACEXTBL  
WHERE Launch_Site LIKE 'CCA%'  
LIMIT 5;
```

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with 'CCA'
- Retrieved 5 launch records with launch sites starting with CCA.
- Most records belong to **CCAFS LC-40** or **CCAFS SLC-40**.
- Demonstrates the importance of Cape Canaveral in SpaceX launch operation
- Present your query result with a short explanation here
- The query returned the first 5 launch records where the launch site name begins with "CCA" (Cape Canaveral Air Force Station).

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
- Present your query result with a short explanation here
- **Result:** The query calculates the total payload mass carried by SpaceX boosters for NASA Commercial Resupply Services (CRS) missions.

```
SELECT SUM(PAYLOAD_MASS_KG_)
FROM SPACEXTBL
WHERE CUSTOMER = 'NASA (CRS)';
```



```
SELECT AVG(PAYLOAD_MASS_KG_)
FROM SPACEXTBL
WHERE BOOSTER_VERSION = 'F9 v1.1';
```

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- Present your query result with a short explanation here
- **Result:** The query calculates the average payload mass carried by the **Falcon 9 v1.1** booster version.
- **Finding:** The Falcon 9 v1.1 booster was capable of carrying several payloads with varying masses. The calculated average provides an estimate of its typical payload capacity across missions

```
SELECT MIN(Date)
FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (ground pad)';
```

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
- Present your query result with a short explanation here
- **Result:** The query identifies the earliest date on which a Falcon 9 first stage successfully landed on a ground pad.
- **Finding:** This milestone marked a major breakthrough in SpaceX's reusable rocket program, demonstrating the

```
SELECT booster_version  
FROM SPACEXTBL  
WHERE landing_outcome = 'Success (drone ship)'  
AND payload_mass__kg_ > 4000  
AND payload_mass__kg_ < 6000;
```

Successful Drone
Ship Landing with
Payload between
4000 and 6000

The query identified all Falcon 9 boosters that successfully landed on a drone ship while carrying a payload between **4,000 kg and 6,000 kg**. These missions demonstrate SpaceX's ability to recover boosters during medium-payload launches, improving reusability and reducing launch costs.

```
SELECT
    CASE
        WHEN Mission_Outcome LIKE '%Success%' THEN 'Success'
        ELSE 'Failure'
    END AS Outcome,
    COUNT(*) AS Total
FROM SPACEXTBL
GROUP BY Outcome;
```

Total Number of Successful and Failure Mission Outcomes



Calculate the total number of successful and failure mission outcomes



The query calculated the total number of successful and failed mission outcomes. The results show that SpaceX missions have a much higher number of successful outcomes than failures, indicating a strong overall launch success rate and improving reliability of the Falcon launch program.

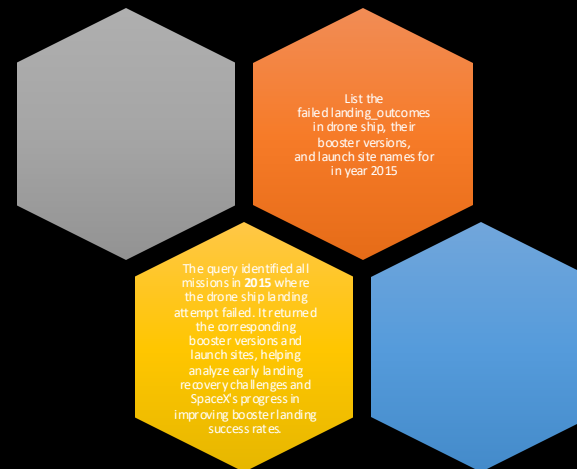
Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- Present your query result with a short explanation here
- The query identified the booster(s) that carried the highest payload mass among all SpaceX missions. These boosters were used for heavy-lift missions, demonstrating Falcon 9's capability to transport large payloads into orbit while maintaining mission success and reusability.

```
SELECT Booster_Version
FROM SPACEXTBL
WHERE Payload_Mass_kg_ = (
    SELECT MAX(Payload_Mass_kg_)
    FROM SPACEXTBL
);
```

```
SELECT Landing_Outcome,  
       Booster_Version,  
       Launch_Site  
FROM SPACEXTBL  
WHERE Landing_Outcome LIKE '%Failure%'  
       AND Landing_Outcome LIKE '%drone ship%'  
       AND Date LIKE '2015%';
```

2015 Launch Records



```
SELECT Landing_Outcome,  
       COUNT(*) AS Outcome_Count  
FROM SPACEXTBL  
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'  
GROUP BY Landing_Outcome  
ORDER BY Outcome_Count DESC;
```

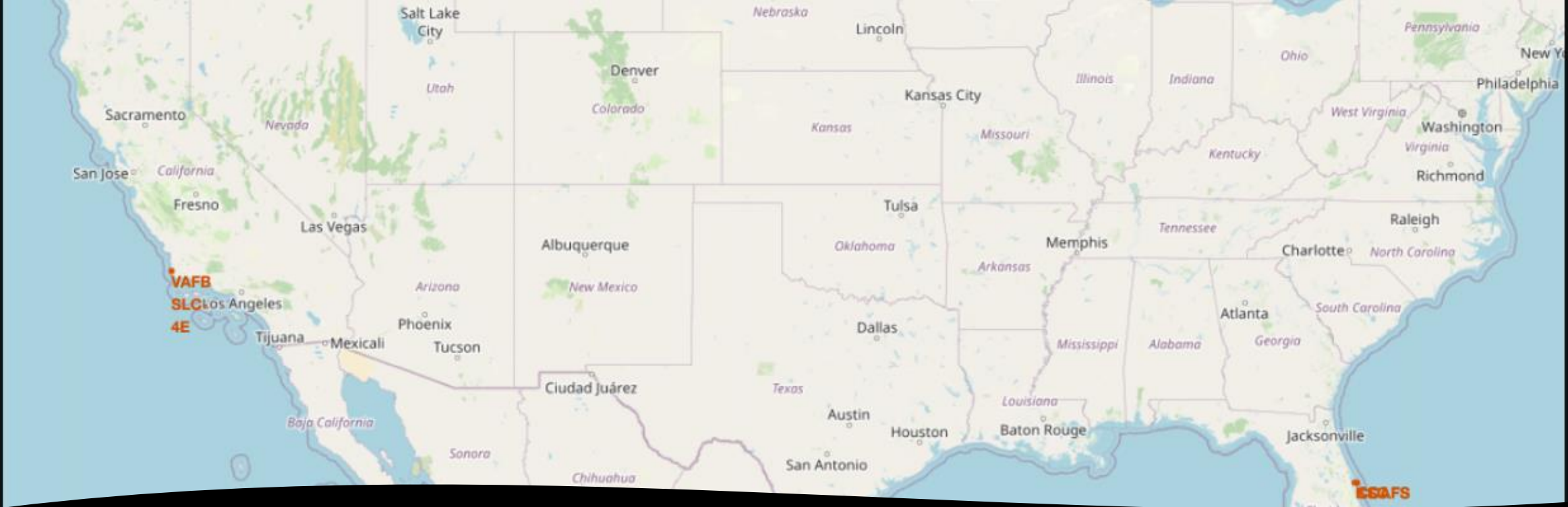
Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Present your query result with a short explanation here
- The query ranks all landing outcomes between **2010-06-04** and **2017-03-20** by their frequency. The results show which landing outcomes occurred most often during this period, providing insight into SpaceX's landing performance and the progression of booster recovery success over time.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue sky on the left and a satellite view of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The lights are concentrated in the lower right portion of the image, following the curve of the Earth. The horizon line of the Earth is visible, separating the dark surface from the blue sky.

Section 3

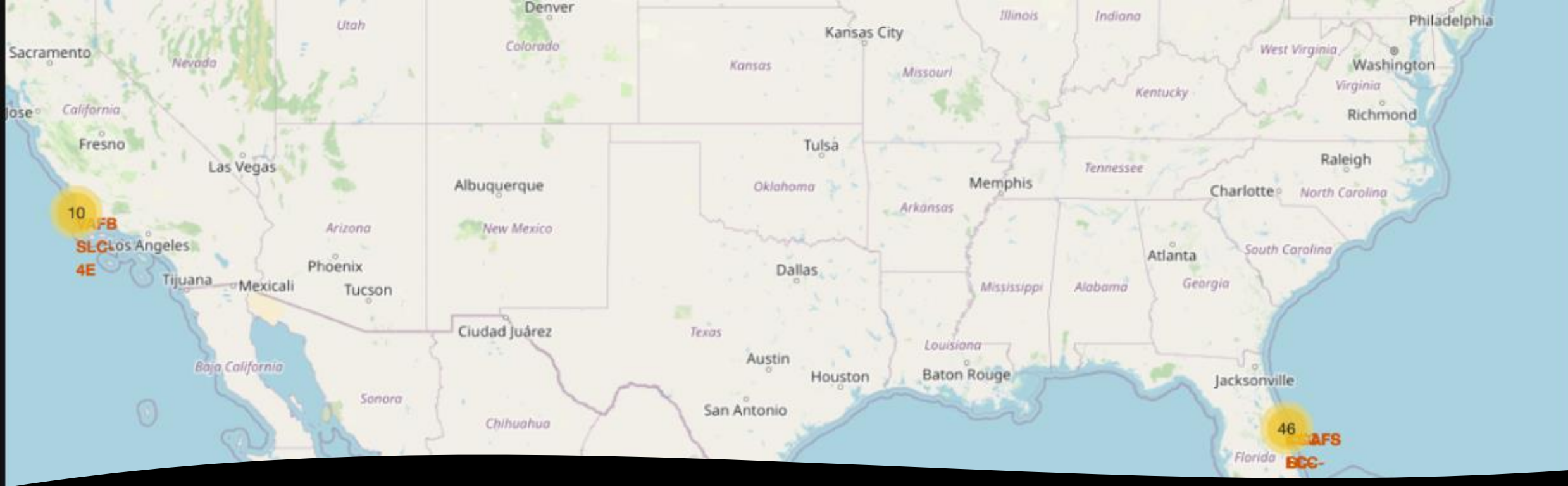
Launch Sites Proximities Analysis



- Explore the generated folium map and make a proper screenshot to include all launch sites' location markers on a global map

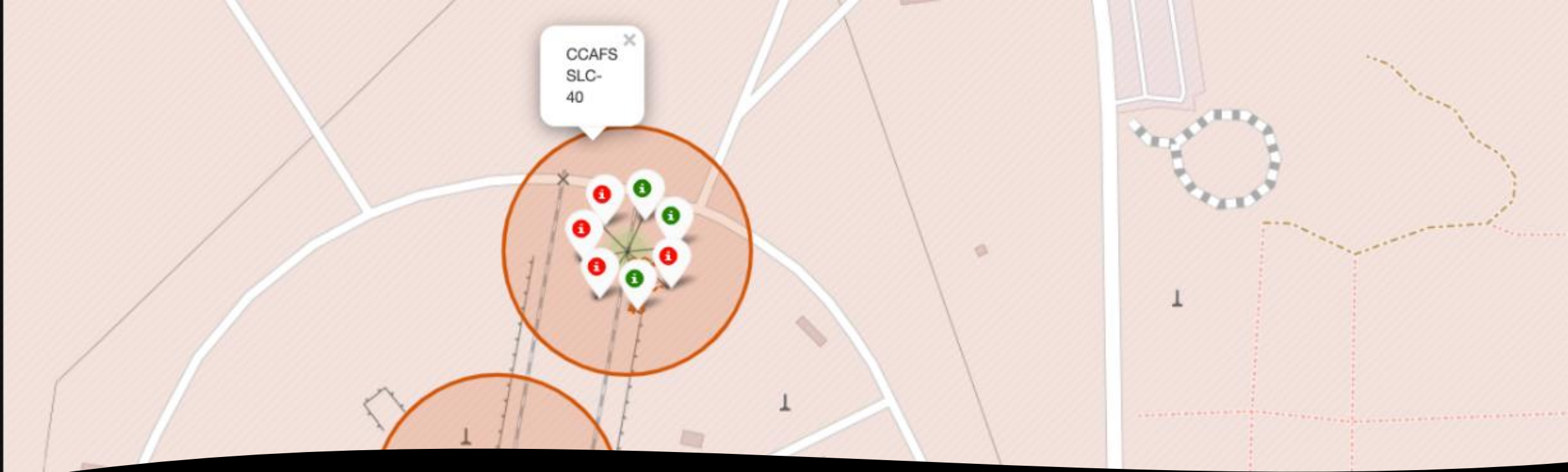
<page 1>

- Now, you can explore the map by zoom-in/out the marked areas , and try to answer the following questions:
- Are all launch sites in proximity to the Equator line?
- Are all launch sites in very close proximity to the coast?



<Page 2>

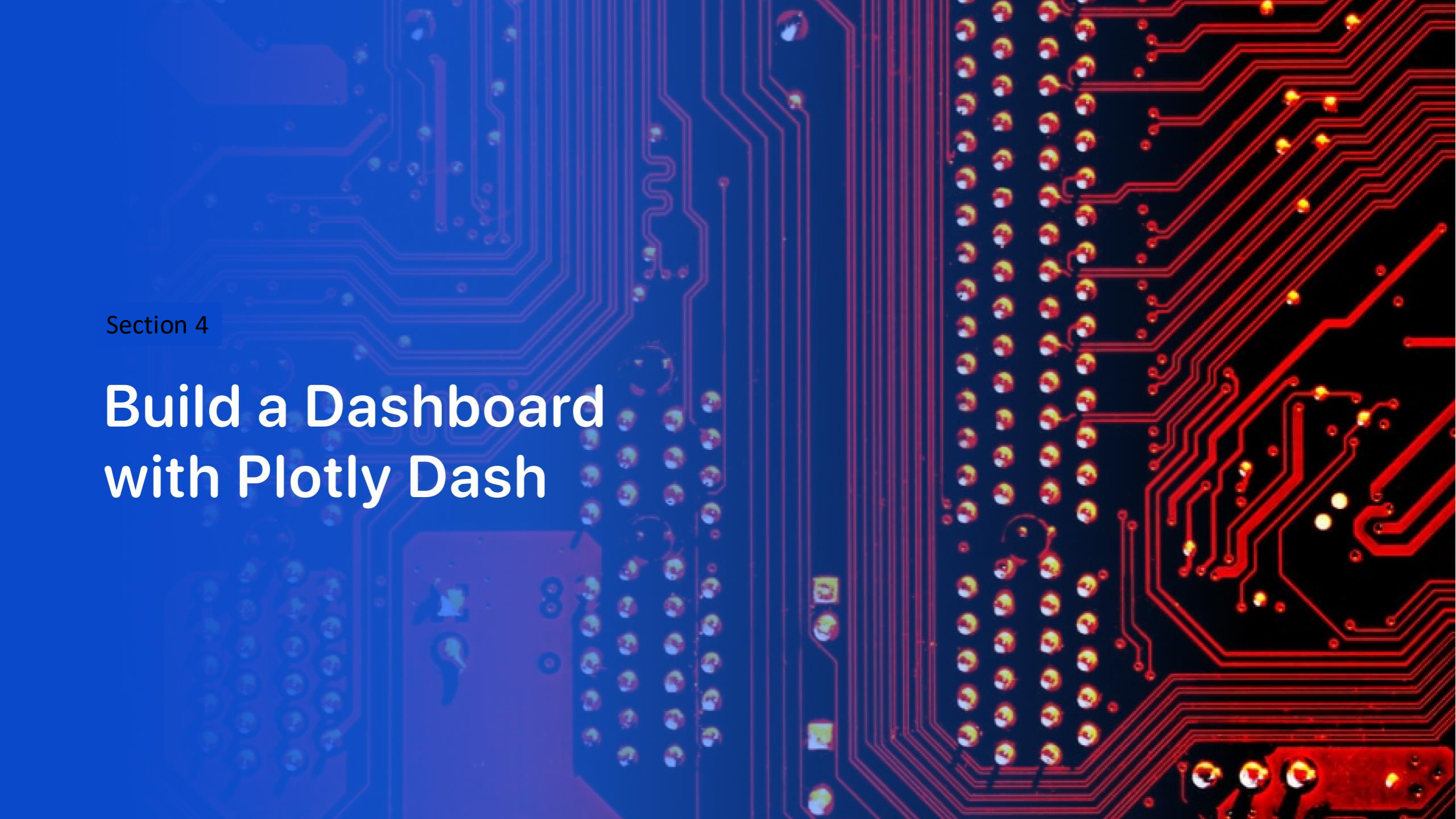
- Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map
- Your updated map may look like the following screenshots:



<page 3>

- Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed
- From the color-labeled markers in marker clusters, you should be able to easily identify which launch sites have relatively high success rates.





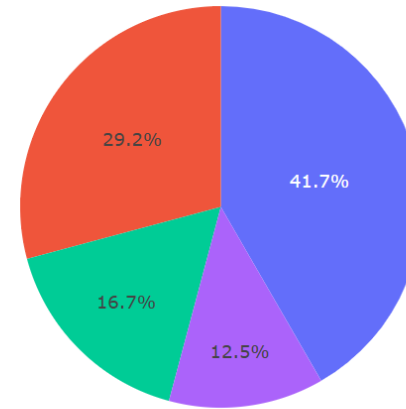
Section 4

Build a Dashboard with Plotly Dash

Dial 1

- Launch Site Dropdown is working
 - ✓ Pie Chart is displaying
 - ✓ Payload Range Slider is displaying
 - ✓ Scatter Plot is displaying
 - ✓ Callback functions are working

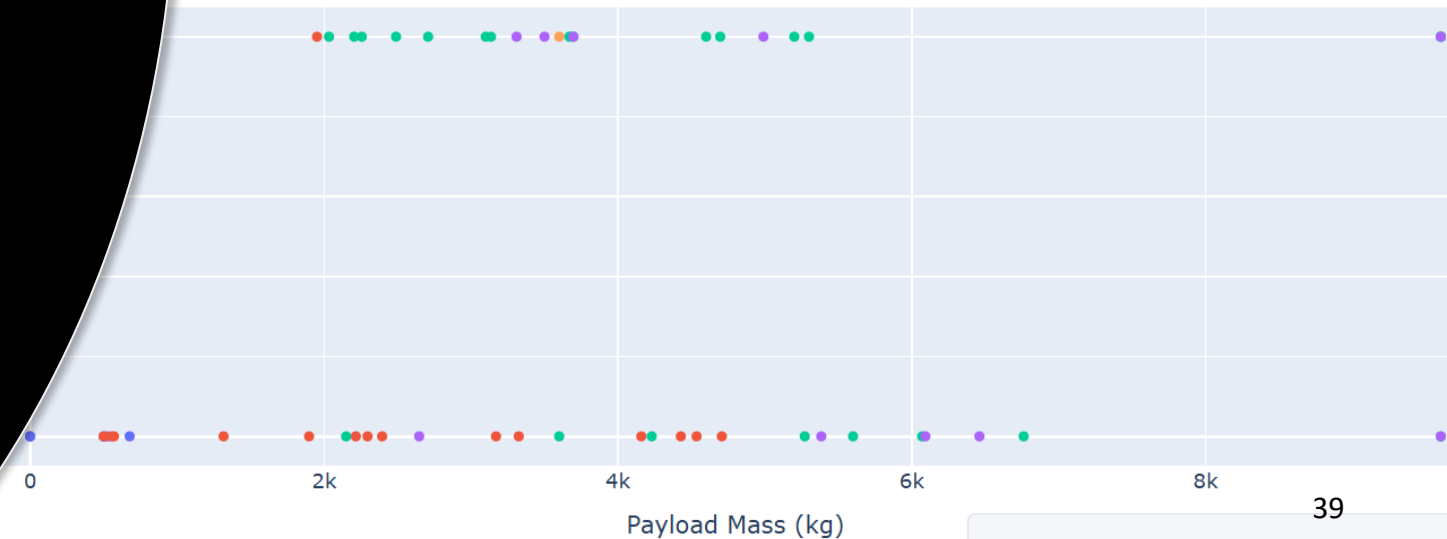
Launch Site



■ KSC LC-39A
■ CCAFS LC-40
■ VAFB SLC-4E
■ CCAFS SLC-40

Plotly Cloud Errors Callbacks | v4.1.0 | Server

Open Payload and Success for All Sites

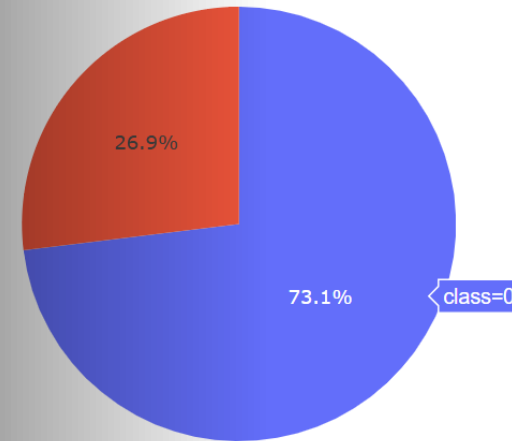


Plotly Cloud Errors Callbacks

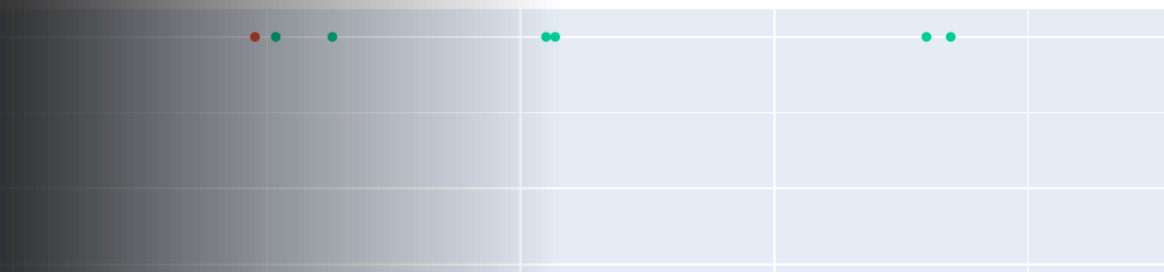
Dial 2

- Pie Chart Analysis
- Success (class = 1): 26.9%
- Failure (class = 0): 73.1%

Failure for CCAFS LC-40

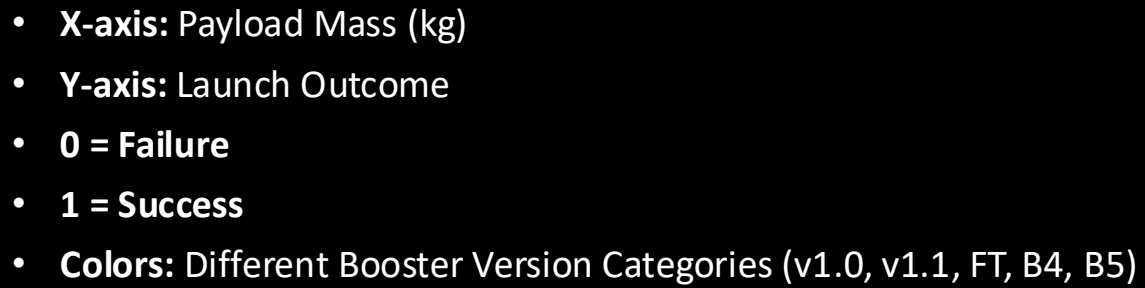


between Payload and Success for CCAFS LC-40



Booster Version Category

- v1.0
- v1.1
- FT

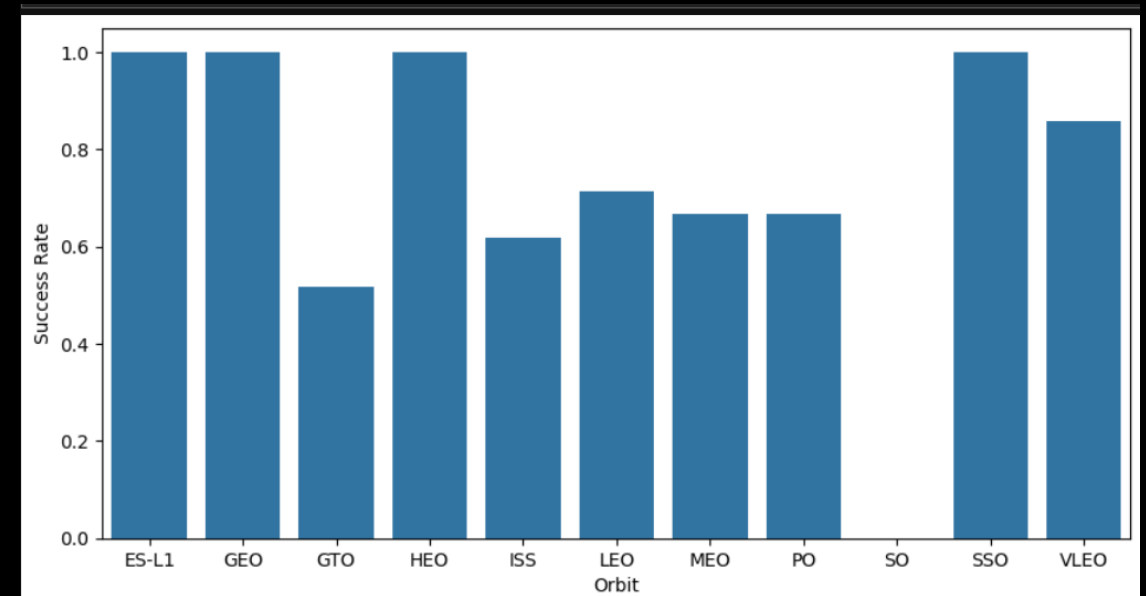


Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart
- Orbit types ES-L1, GEO, HEO, and SSO achieved a 100% launch success rate, indicating excellent mission reliability. GTO showed a lower success rate (~51%), while SO had no successful launches. This suggests that launch success varies significantly across orbit categories.



Confusion Matrix

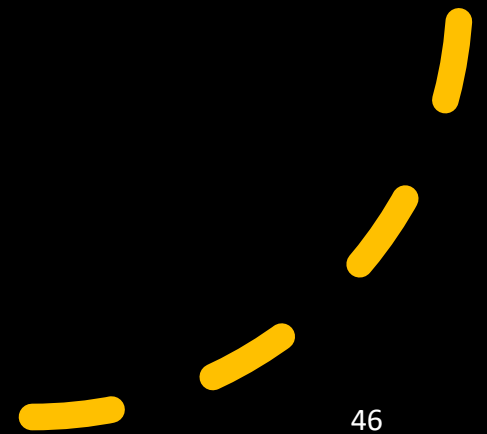
- The confusion matrix evaluates the classification performance of the model by comparing actual outcomes with predicted outcomes.
- **True Positives (TP):** Successful landings correctly predicted as successful.
- **True Negatives (TN):** Failed landings correctly predicted as failures.
- **False Positives (FP):** Failed landings incorrectly predicted as successful.
- **False Negatives (FN):** Successful landings incorrectly predicted as failures.



Conclusions

- **Launch Success Improved Over Time**
 - Falcon 9 launch success rates increased significantly with higher flight numbers, demonstrating improved reliability and operational experience.
- **Payload Mass Influences Mission Success**
 - Successful landings were more common within certain payload mass ranges, indicating a relationship between payload weight and landing outcomes.
- **Launch Site and Orbit Matter**
 - Different launch sites and orbit types showed varying success rates. GEO, HEO, ES-L1, and SSO missions achieved the highest success rates.
- **Predictive Modeling Was Effective**
 - Machine learning models successfully predicted first-stage landing outcomes. The best-performing model achieved high classification accuracy and can support future mission planning.
- **Business Impact**
 - Accurate landing predictions help estimate launch costs, improve mission planning, and demonstrate the economic benefits of reusable rocket technology.
- **Overall Finding**
 - SpaceX's reusable Falcon 9 system has achieved high reliability, and data-driven analysis can effectively predict landing success for future launches.

Appendix



Thank you!

